

API USER GUIDE

1. Structures

1.1 ALGPDS_INFO_S

[Description]

Information of pedestrian detection result.

[Syntax]

```
typedef struct ALG_PDS_INFO_S
{
    int num;
    ALGPDS_RECT_S Rect_t[50];
}ALGPDS_INFO_S;
```

[Member]

Member	Description
num	How many items are valid in array 'Rect_t'
Rect_t	Detail information about each detected box

[Note]

The data of this structure comes from function ALGPDS_get_result.

You should check the member 'num' , and get box detail form 'Rect_t' by index [0, ... , num] .

1.2 ALGPDS_RECT_S

[Description]

Detail information about each detected box.

[Syntax]

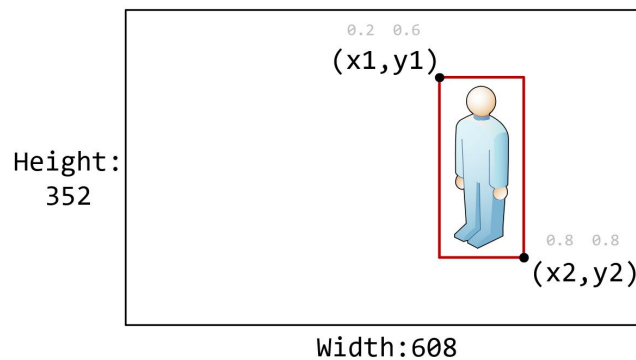
```
typedef struct ALGPDS_RECT_
{
    float x1;
    float y1;
    float x2;
    float y2;
    float confidence;
    float distance;
    float reserve;
    unsigned short classes;
}ALGPDS_RECT_S;
```

[Member]

Member	Description
x1	The x coordinate of the left top corner of the box
y1	The y coordinate of the left top corner of the box
x2	The x coordinate of the right bottom corner of the box

y2	The y coordinate of the right bottom corner of the box
confidence	The percentage that you can consider there is really a target in the box
distance	Not used now
reserve	Not used now
classes	The index of class for this box

[Note]



Noted that the coordinate of box is **normalized**. For example, your original image is 608x352, and the Deep Neural Network recognized a person in the image, then tell you that x1 is 0.2 and y1 is 0.6, so the actual coordinate in original image is $x1=0.2 \times 608$, $y1=0.6 \times 352$.

2. Functions

2.1 ALGPDS_LibVersion

[Description]

The version of algorithm library specified by developer.

[Syntax]

```
const char* ALGPDS_LibVersion();
```

[Parameter]

Parameter	Description	Input/Output

[Return Value]

Return Value	Description
version	A pointer to string

2.2 ALGPDS_ModelVersion

[Description]

The version of neural network model (RKNN) specified by developer.

[Syntax]

```
const char* ALGPDS_ModelVersion();
```

[Parameter]

Parameter	Description	Input/Output

[Return Value]

Return Value	Description
version	A pointer to string

2.3 ALGPDS_init

[Description]

Initialize the algorithm library.

[Syntax]

```
int ALGPDS_init(float *thresholds, int numberOfThresholds, char* pModelFile);
```

[Parameter]

Parameter	Description	Input/Output
thresholds	The pointer of input thresholds array. [0] :confidence threshold [1] :NMS threshold [2] :VIS threshold	Input
numberOfThresholds	Numbers of input thresholds	Input
pModelFile	Model file path	Input

[Return Value]

Return Value	Description
0	Success

2.4 ALGPDS_forward

[Description]

Run the neural network algorithm.

[Syntax]

```
int ALGPDS_forward(char* p_inputdata);
```

[Parameter]

Parameter	Description	Input/Output
p_inputdata	RGB data of image. Please make sure that the size of data is the same to network input.	Input

[Return Value]

Return Value	Description
0	Success

2.5 ALGPDS_get_result

[Description]

Decode data from neural network and get final result.

[Syntax]

```
int ALGPDS_get_result(ALGPDS_INFO_S *pResult);
```

[Parameter]

Parameter	Description	Input/Output
pResult	Details of detection result.	Output

[Return Value]

Return Value	Description
0	Success

2.6 ALGPDS_release

[Description]

Release all resources.

[Syntax]

```
int ALGPDS_release();
```

[Parameter]

Parameter	Description	Input/Output
-----------	-------------	--------------

[Return Value]

Return Value	Description
0	Success

3. Procedure

